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University of Ha'il

College of Computer Science and Engineering
Students Projects Committee

Graduation Design Project Proposal Form

Of

[CityShield Platform: Simulating and Defending Smart City Cyberattacks]

By

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Submitted to the [Information Security] Department in partial fulfillment of the requirements for the semester course Graduation Project | ([CityShield Platform: Simulating and Defending Smart City Cyberattacks])

Supervised by:
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Project Title: **CityShield Platform: Simulating and Defending Smart City Cyberattacks**

Project Type: Graduation Design Project

Proposed by: Sami Alsulaymi, Khaled Alshammari, Abdulwahab Alfayez, Thamer Alshammari, and Bati Albata

The enhanced CityShield Platform is designed to simulate and defend smart city cyberattacks within a controlled and secure environment. It introduces several architectural and functional improvements to increase realism, scalability, and educational value.

The platform defines three primary user roles to ensure secure access control and collaboration across research and analysis teams:

- **Administrator:** Manages system infrastructure, user access permissions, and deployment of city microservices.
- **Analyst:** Monitors dashboards, investigates alerts, and validates incident responses within the SIEM environment.
- **Researcher:** Conducts safe attack simulations, studies results, and proposes new defensive mechanisms.

Additionally, the platform integrates live threat intelligence feeds to enrich alerts with contextual information such as IP reputation, indicators of compromise (IOCs), and emerging attack trends.

The system also provides RESTful APIs for modular integration, enabling researchers and developers to connect external tools, extend detection models, or add visualization modules.

Through these enhancements, CityShield evolves into a scalable and interactive cyber defense range that supports both practical experimentation and academic research in smart city cybersecurity.

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Aim: To build a secure, scalable, and interactive simulation environment that enables safe experimentation with cyber-attacks and defenses in smart cities.

Project Objectives:

- Develop modular microservices representing city domains.
- Build a Smart City.
- Simulate realistic cyber-attack scenarios.
- Deploy SIEM rules and ML-based anomaly detection.
- Implement automated response playbooks.
- Provide a visual dashboard and replay of incidents.

The main objective to the proposed system: build a secure and interactive smart city simulation environment that enables safe testing of cyberattacks and evaluation of defense mechanisms.

Methodology:

Architecture



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The system architecture is based on containerized Docker Compose or Kubernetes clusters, providing isolation and scalability for each service. Every smart city domain—such as electronic health records, traffic control, IoT smart meters, payment processing, and CCTV surveillance—is deployed as an independent microservice. All generated data flows through a centralized pipeline: File beat → OpenSearch → Kibana, which enables unified monitoring, indexing, and visualization. To maintain flexibility and extensibility, the platform exposes RESTful APIs that allow third-party tools, visualization modules, or AI models to integrate seamlessly into the existing infrastructure.

Detection

The detection mechanism combines traditional rule-based analytics with advanced intelligence-driven models. Core detection logic is implemented through ElastAlert rules, aligned with the MITRE ATT&CK framework to visualize attack paths, techniques, and procedures across different domains. Machine learning models detect behavioral anomalies in user or network activity, helping to identify zero-day or previously unseen attacks. Furthermore, live threat intelligence integration enriches each alert with contextual threat data, improving accuracy while reducing false positives and false negatives.

Response

Automated playbooks built using Ansible and Python handle incident containment and mitigation. When an alert is triggered, the system can automatically isolate compromised assets, revoke access tokens, or block suspicious endpoints. Response performance is continuously evaluated to reduce human intervention and improve Mean Time to Respond (MTTR).

Evaluation Metrics

To ensure measurable performance and continuous improvement, several cybersecurity metrics will be tracked throughout the implementation and testing phases:

- **Mean Time to Detect (MTTD):** Measures how quickly the system identifies an active threat.
- **Mean Time to Respond (MTTR):** Measures the time required to contain and neutralize an attack.
- **Detection Accuracy:** Evaluates the ratio of true positives to total detections.
- **False Positive Rate:** Monitors incorrect or misleading alerts to assess detection precision.
- **Resource Utilization:** Analyzes CPU, memory, and network usage under different workloads to ensure scalability.

Expected Learning outcomes:

Upon successful completion of the project, the team members are expected to achieve the following learning outcomes:

- Gain **hands-on experience** in building, securing, and monitoring smart city environments.
- Understand the **process of simulating and defending cyberattacks** within a controlled infrastructure.
- Acquire **practical skills** in SIEM configuration, log correlation, and ML-based anomaly detection.
- Develop proficiency in **automating incident response workflows** using Ansible and Python playbooks.
- Enhance **collaboration, documentation, and presentation** skills through structured teamwork.
- Utilize **educational dashboards** that provide interactive visualizations of attack paths, response times, and detection accuracy—serving as a valuable training tool for both academic and professional cybersecurity programs.



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Expected project outcomes:

- A working smart city cyber range.
- Attack-defense simulation scripts.
- Configured dashboards for real-time monitoring.
- Dataset of labeled attack/defense events.
- Final report and technical documentation .

Resources Needed:

- **Software Resources:** Docker, Kubernetes (k3s), OpenSearch/Elasticsearch, Kibana, Filebeat, Python, Ansible, Grafana.
- **Hardware Resources (If applicable):** Lab server or VM host with ≥ 16 GB RAM and ≥ 4 CPU cores.

Software tutoring/help to be provided by the Lab Instructor:

- Support with Docker and OpenSearch setup.
- Guidance on safe IoT simulation and network segmentation.
- Assistance with integrating anomaly detection models.

Identified tasks and a tentative work plan [[Mention all major tasks/sub-tasks to be accomplished.](#)]

Deliverables	Task	Expected End Date
Architecture report.	Conduct research and design the overall system architecture.	1-2
Functional prototypes.	Develop core microservices for EHR, IoT, and traffic systems	3-4
Log aggregation and monitoring setup	Integrate SIEM components and establish the data collection pipeline.	5-6
Operational detection engine.	Develop ML-based anomaly detection and alerting mechanisms.	7-8
Completed automation scripts.	Implement automated response playbooks and test their functionality.	9-10
Performance metrics (MTTD, MTTR, accuracy).	Evaluate system performance and optimize detection and response.	11



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Final report and presentation slides.	Finalize documentation, testing, and prepare the presentation.	12
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Team Information:

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References: